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Building a Multimodal Dashboard with Large Language Models and Chatbot Integration for Interactive Data Analysis

Final Presentation - Challenges in Supply Chain Management - Group 3

Agenda

- Motivation: Why do we need a multimodal chatbot?
- Starting Point: The initial idea of our chatbot
- Mid-Journey: Idea pipelines and related learnings
- Result: A quick overview of our final result
- Result: Demo of our final result
- Outlook: Limitations and possible next steps

A chatbot-based dashboard enables efficient, accessible, and round-the-clock data management, scaling with company growth

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Problems that occur with current dashboards

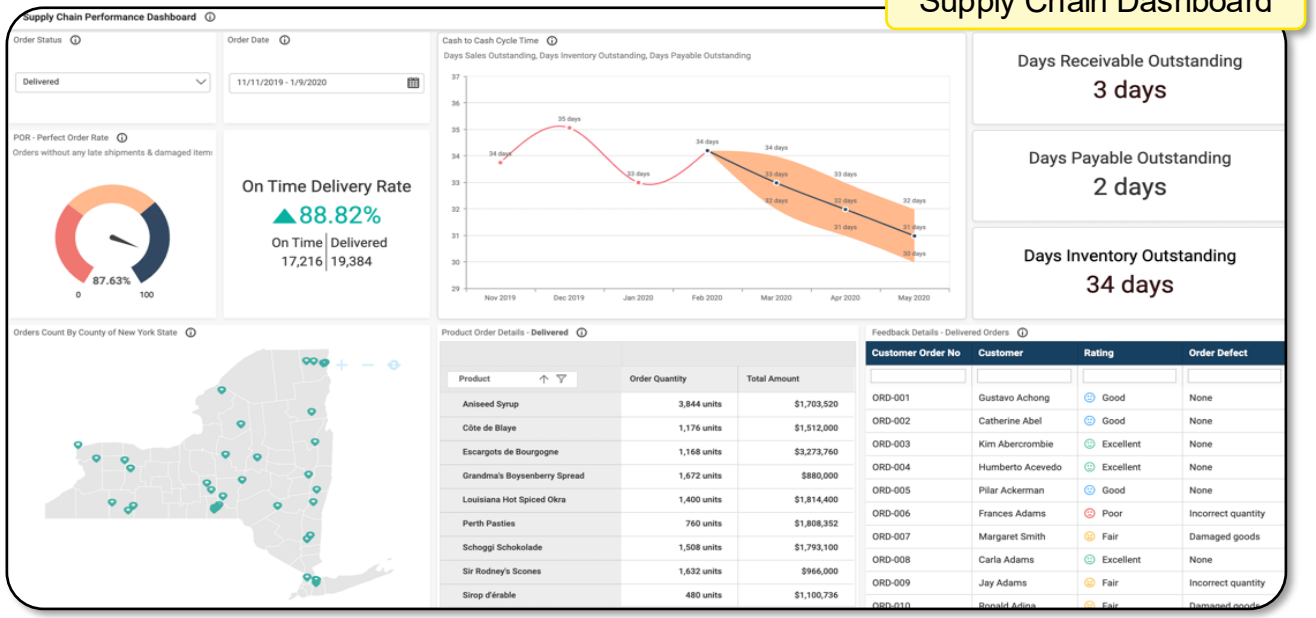
- Complex dashboard show low usability
- High Data Overload
- Time-consuming data retrieval
- Lack of real-time interaction

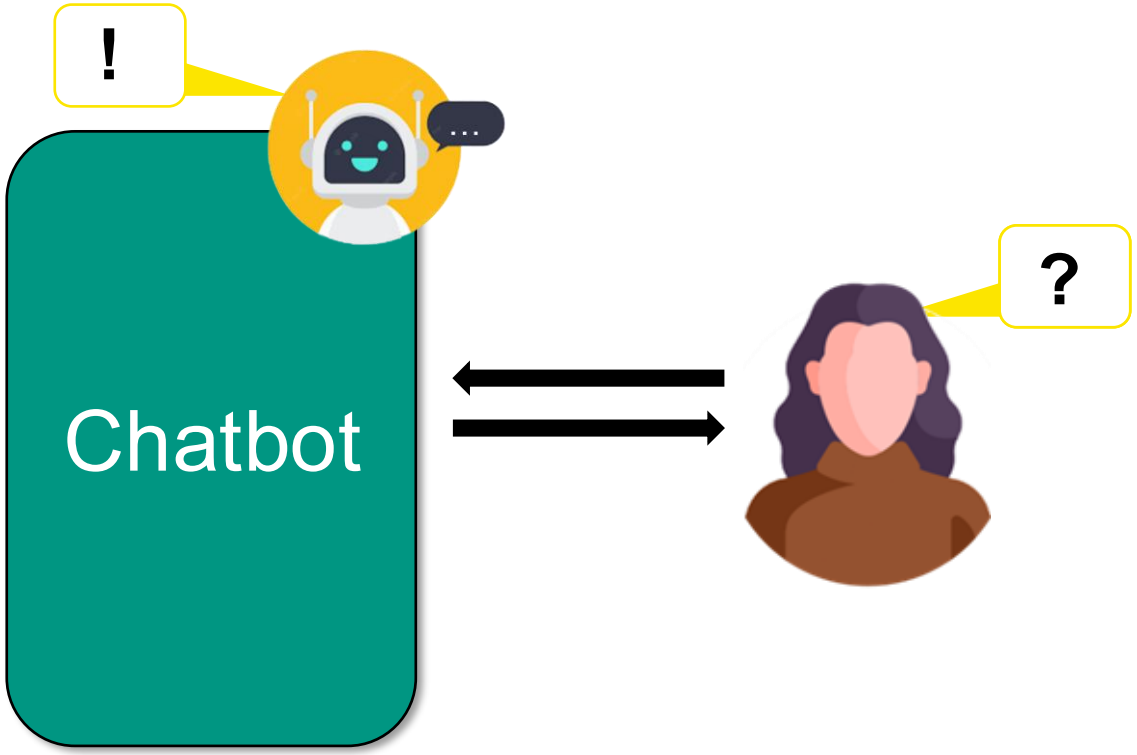
Solutions that a chatbot based dashboard offers

- Chatbot as a single source for all queries
- Reduction to query based data display
- High efficiency of LLM based data retrieval
- 24/7 availability of chatbot interactions

Starting point: Augment an existing dashboard with a chatbot for improved data point readout and interactive data analysis

Supply Chain Dashboard





Since the original idea implementation, a lot of work has been put into pivoting between various chatbot alternatives.

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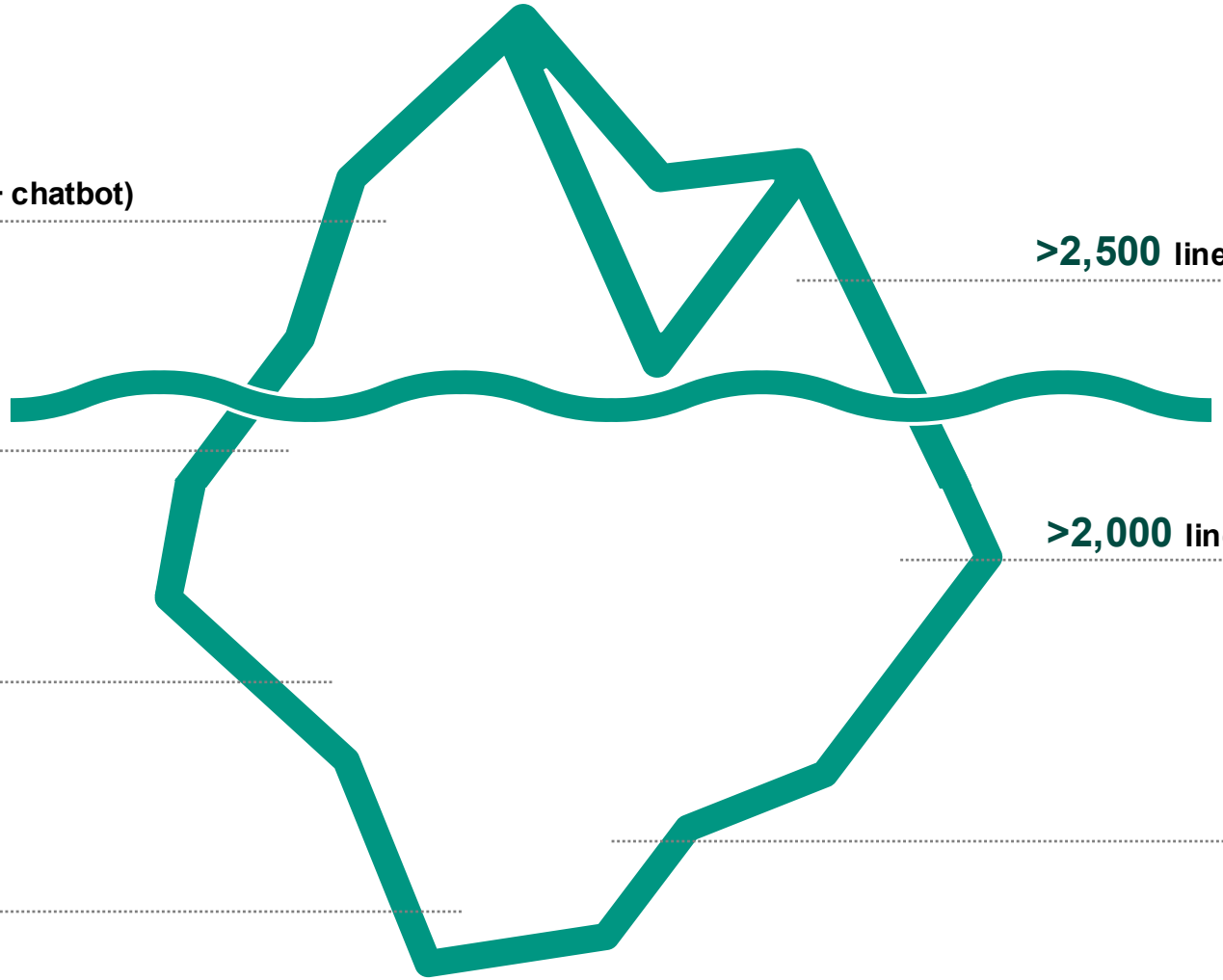
1 final artefact (dashboard + chatbot)

3 previous artefacts

300 hours of coding

20 pages of documentation



>2,500 lines of code in final artefact

>2,000 lines of code in old artefacts

6 Key Learnings

On our journey multiple pivots and change of plans were involved. We want to share an excerpt of our main learnings.

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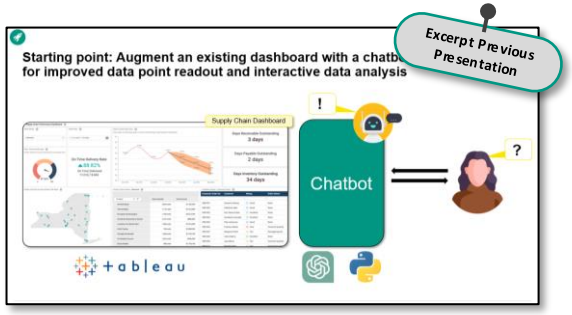
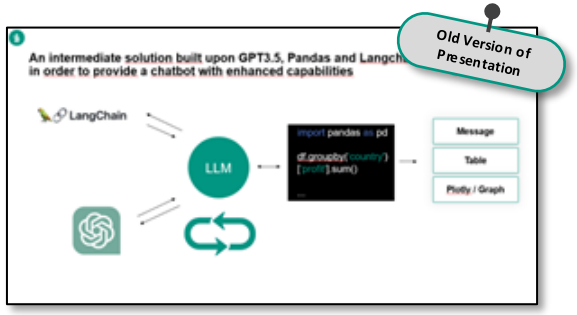
Pipeline 1: Tableau



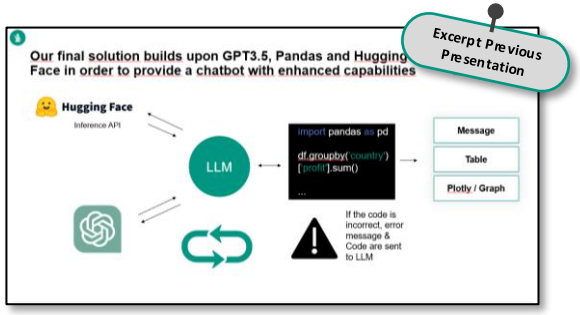
The initial idea was based on introducing a chatbot to tableau using existing extension.

Pipeline 2: Langchain



A revised idea was using Langchain to interact with data and provide Plotly commands.

Pipeline 3: Own Artefact



Our final artefact uses i.a. GPT4, Plotly & Pandas to provide enhanced chatbot capabilities.

1. **Tableau provide restricted features for data interaction for the chatbot**
2. **Tableau will publish their version of an LLM integration in the near future**

3. **The documentation of Langchain is very detailed and great as a start**
4. **Not having full access to Langchain features, limits chatbot capabilities**

5. **Setting up an own chatbot gives freedom for endless features ...**
6. **… at the cost of heavily increased implementation time**

Our final solution builds upon GPT4, Pandas and Plotly in order to provide a chatbot with enhanced capabilities

LLM

5x

Python Engine

```
import pandas as pd

df.groupby('country')
['profit'].sum()

...
```

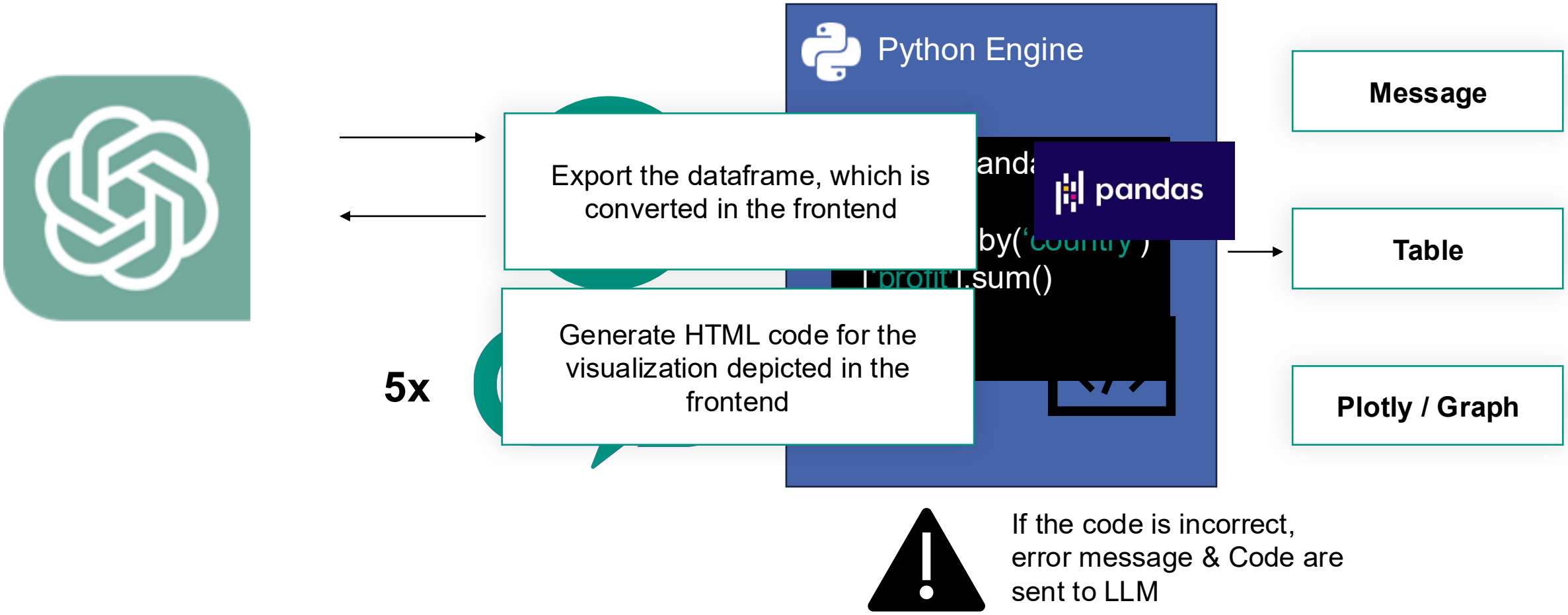
Message

Table

Plotly / Graph

If the code is incorrect, error message & Code are sent to LLM

Our final solution builds upon GPT4, Pandas and Plotly in order to provide a chatbot with enhanced capabilities

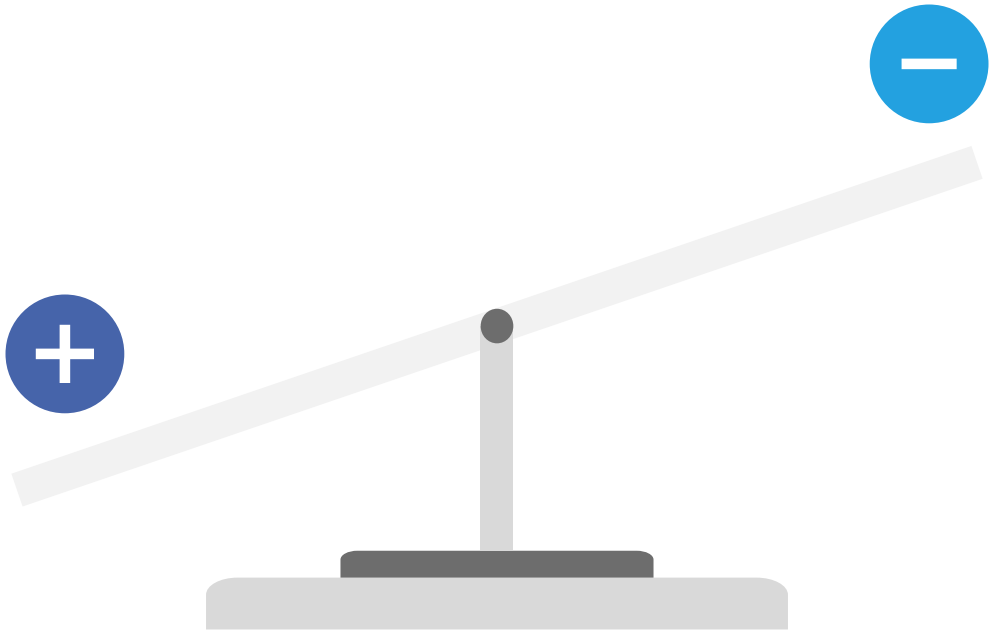


Why chaining several iterations with GPT4?

Significantly better performance
GPT4 gets direct feedback and can automatically adjust the code

Dynamic Adaptions
Dynamically adapt the requirements

Modular extensible
Additional chains can be added very easily



Increased Latency
Several chained API calls lead to longer waiting periods

Higher Cost
More prompts/tokens are sent to GPT4

Loops
Sometimes solutions to an error are mutually dependent

Next, we want to have a look on the final version of our chatbot. Here we go:

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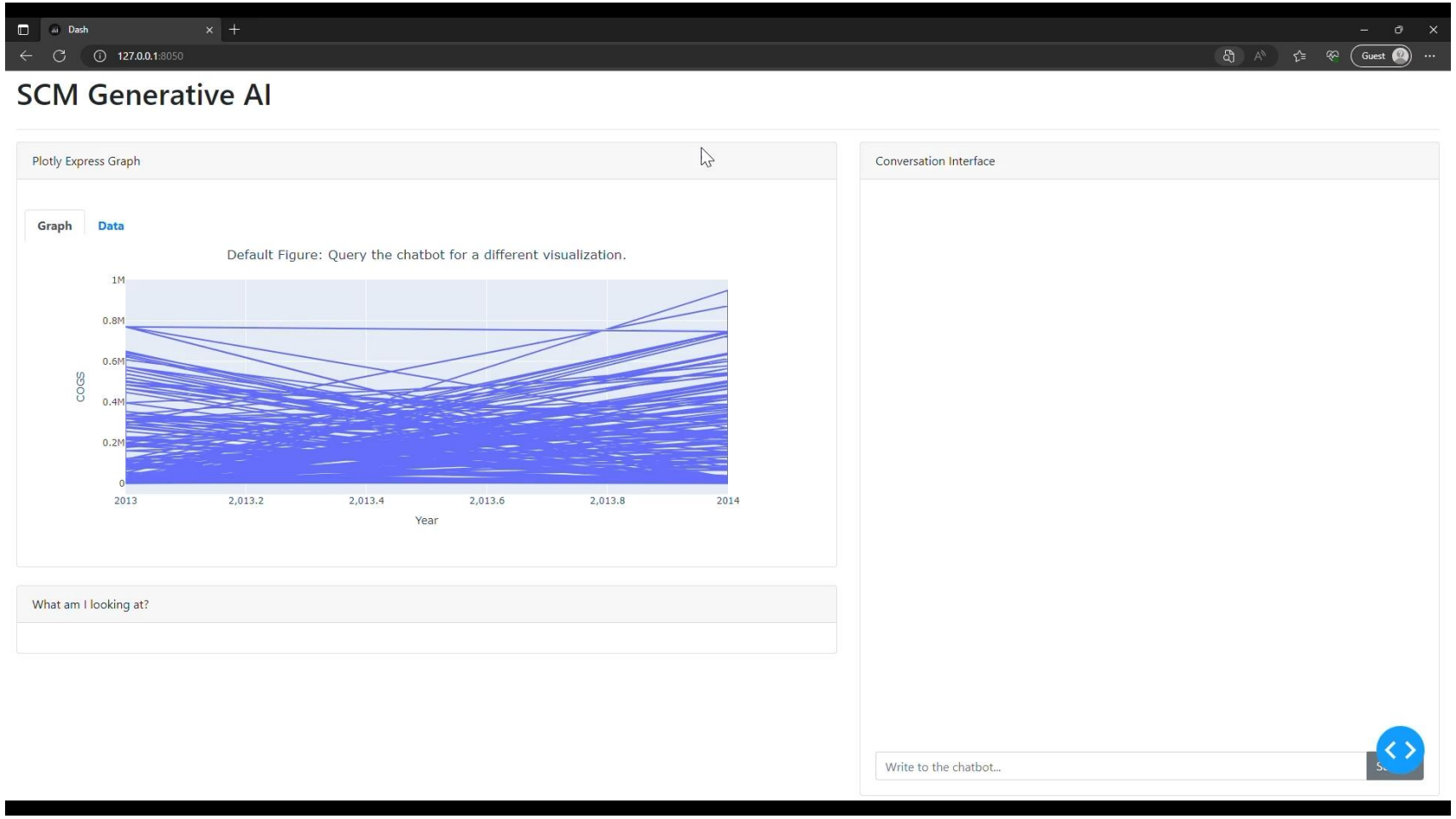
Demo



Viable next steps describe applying our chatbot on further use cases and testing other, finetuned, LLMs for improved results

1

Implementation on various BASF use cases

- Thus far, we have built the dashboard for sales data. The use of other datasets and the inclusion of more complex graphical applications are of interest.

2

Performance testing on other LLMs

- Our dashborad relies on OpenAI‘s GPT models. The introducing of self-owned LLM could be an viable path to achieve higher data security.

3

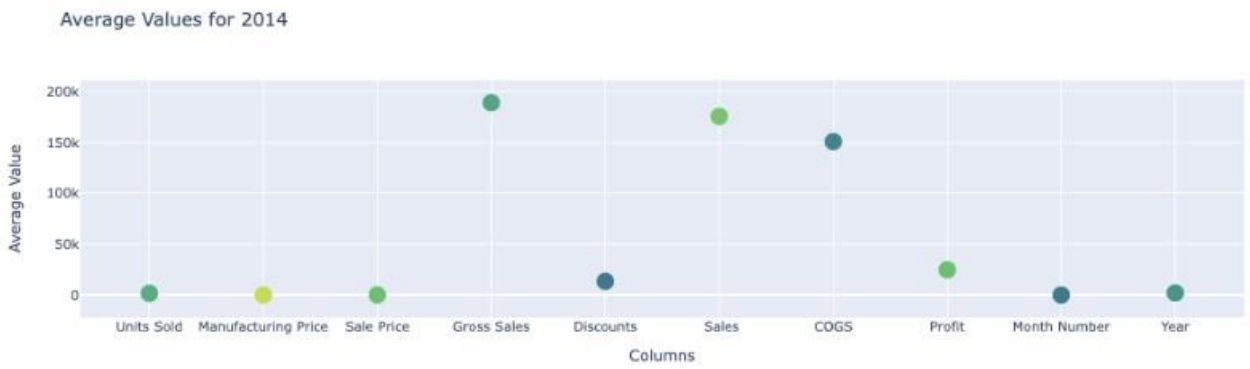
Finetuning of the underlying LLM

- We also perceive a BASF data finetuned LLM as a valuable augmentation to our artefact for even enhanced dashboard performance

Thank you for your attention!

Any Questions?

Graph Data



What am I looking at?

Code used for calucalting the values:

```
# Filter the dataframe for the year 2014
df_2014 = df[df['Year'] == 2014]

# Calculate the averages for numerical columns in the filtered dataframe
# Exclude non-numeric columns by selecting only numeric data types
average_2014 = df_2014.select_dtypes(include=[np.number]).mean()

print(average_2014)
```

Code used for the visualization:

```
import plotly.graph_objects as go
import plotly.io as pio
import numpy as np

df_2014 = df[df['Year'] == 2014]
average_2014 = df_2014.select_dtypes(include=[np.number]).mean()

fig = go.Figure(data=[go.Scatter(name='Average', x=average_2014.index, y=average_2014.values, mode='markers', marker=dict(color=np.random.randn(500), size=
fig.update_layout(title_text='Average Values for 2014', xaxis_title='Columns', yaxis_title='Average Value', height=425)

print(pio.to_json(fig))
```

What averages and filter for 2014?

CATEGORY	VALUE
Units Sold	1640.2514285714
Manufacturing Price	96.419047619
Sale Price	118.4285714286
Gross Sales	189277.7361904762
Discounts	13447.0795238095
Sales	175830.6566666667
COGS	151039.7276190476
Profit	24790.929047619
Month Number	7.0666666667
Year	2014

Write to the chatbot... Submit

SCM Generative AI

Plotly Express Graph

Graph Data

What am I looking at?

Code used for calucalting the values:

```
germany_df = df[df['Country'] == 'Germany']
correlation = germany_df['Gross Sales'].corr(germany_df['Profit'])
print(correlation)
```

Code used for the visualization:

```
import plotly.graph_objects as go
import plotly.io as pio

germany_df = df[df['Country'] == 'Germany']
fig = go.Figure(data=go.Scatter(x=germany_df['Gross Sales'], y=germany_df['Profit'], mode='markers',
                                marker=dict(size=10, color='rgba(255, 182, 193, .9)', line=dict(width=2, color='rgba(0, 0, 0, .8)'))))
fig.update_layout(title='Correlation between Gross Sales and Profit for Germany', xaxis_title='Gross Sales',
                  yaxis_title='Profit', height=400)
print(pio.to_json(fig))
```

Conversation Interface

Correlate Gross Sales and Profit but only for germany

0.8221920807737894

Write to the chatbot...